INITIAL DEFENCE PRESENTATION

COMPUTER SCIENCE DEPARTMENT

Quantum-Inspired Network Intrusion Detection System

Optimizing ML-based NIDS using Quantum-Inspired Algorithms for Feature Selection & Classification

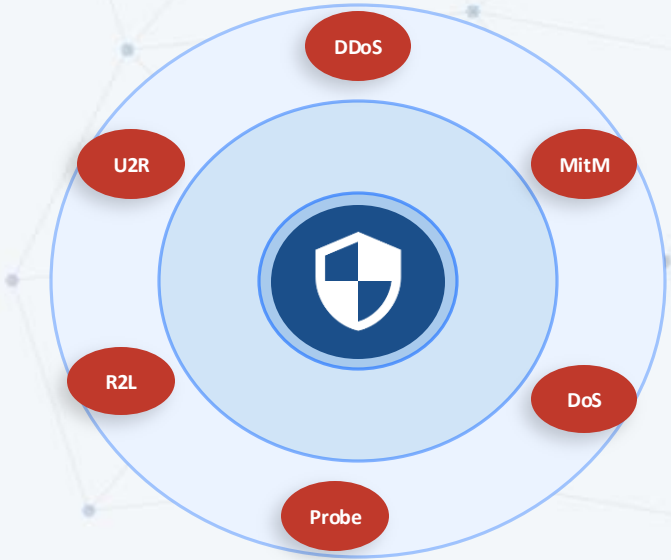
Presented by	Roll No	Supervisor	Session
Anita Bashir	**92**	**Prof. Dr Syed Ali Abbas**	**2022–2026**
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PROBLEM STATEMENT

Why Do We Need This System?

Network intrusion is a growing threat — existing solutions fall short



Global

- $10.5 Trillion cybercrime cost by 2025
- Cyberattacks ↑ 15% every year
- Traditional NIDS = high false-positive rates

PK Nationally

- Banks & telecom sectors under attack
- No indigenous cybersecurity tools
- Real-time detection gap in critical infra

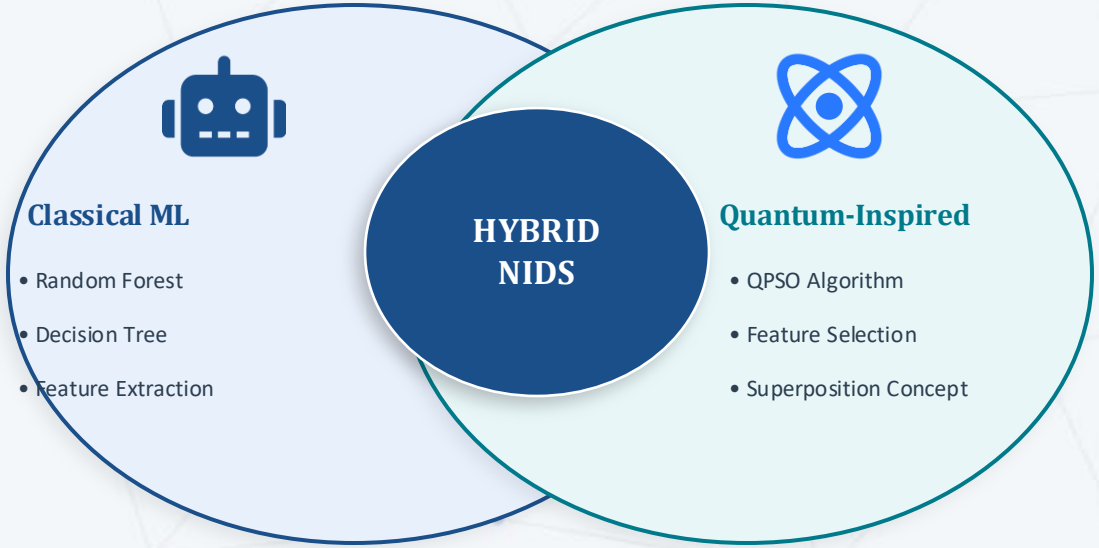
Local

- Campus networks unprotected
- Manual monitoring — inefficient
- Research gap: quantum-ML NIDS not explored

PROPOSED SOLUTION

Hybrid ML-NIDS with Quantum-Inspired Optimization

Combining classical ML with QPSO for smarter, faster intrusion detection



Smart Feature Selection

41 features → 18-22 via QPSO

Higher Accuracy

~88% → ~94%+ detection rate

Lower False Positives

12% FPR → ~5% FPR

GUI Simulator

Visual dashboard + report

METHODOLOGY 1 / 3

Data Input & Preprocessing

① Data Input & Preprocessing | ② ML Model Training & QPSO | ③ Detection & GUI

Data Input

- NSL-KDD Dataset & UNSW-NB15 Dataset
- Real-world network traffic logs

Data Preprocessing

- Normalization — Min-Max Scaling · Z-Score
- Remove noise, nulls & duplicates

Feature Engineering & Split

- Extract 41 initial features from traffic data
- 80% Train / 20% Test — Stratified split

Datasets Used

- NSL-KDD (UNB.ca)
- UNSW-NB15 (ADFA)

Output → Clean, Normalized, Feature-ready Dataset — Passes to Phase 2

ML Model Training & QPSO Optimization

Classical ML Model Training

- Random Forest (100 trees) & Decision Tree (CART)
- Train on dataset → Save Baseline Metrics

QPSO Feature Selection

- Initialize particles (each = feature subset), evaluate fitness
- Quantum velocity update → converge to best 19 features

Hybrid RF + QPSO Model

- Re-train Random Forest on 19 optimized features
- Evaluate: Accuracy · Precision · Recall · F1 · FPR

Quantum-Inspired Opt.

- Particle Swarm Algorithm
- Fitness Evaluation
- Convergence Detection

QPSO Algorithm

- Dimensionality Reduction
- Optimal Feature Subset
- 41 → 19 features (53% ↓)

Output → Hybrid QPSO+RF Model Trained · Baseline Metrics Saved → Passes to Phase 3

Attack Detection & GUI Simulation

① Data Input & Preprocessing | ② ML Model Training & QPSO | **③ Detection & GUI**

Attack Detection & Classification

- Apply Hybrid RF model — Attack / Normal classification
- Confidence Scoring & Alert Generation

Classical vs Quantum Comparison

- Compare baseline RF vs QPSO+RF metrics
- Validate improvement in Accuracy, FPR & F1-Score

GUI Visualization & Reporting

- Real-Time Dashboard — live detection, metrics & charts
- Export Results — PDF / CSV Report

GUI Features

- Live Attack Feed
- Metrics Dashboard
- Classical vs QPSO View
- Custom Attack Injection

Complete QPSO-Enhanced NIDS — Detection · Analysis · Visualization

CONCLUSION

What This Project Achieves

A novel Quantum-Inspired approach to Network Intrusion Detection — making cybersecurity smarter

Benchmark Datasets

Leverages two internationally recognized datasets — NSL-KDD and UNSW-NB15 — ensuring reproducible and credible evaluation

Quantum-Inspired Feature Selection

QPSO algorithm reduces 41 raw features to an optimal subset — removing noise and improving model focus on meaningful signals

Hybrid ML Architecture

Combines Random Forest with QPSO optimization — bridging classical machine learning with quantum-inspired intelligence

Real-World Simulation

A Python GUI simulator enables live attack detection visualization and side-by-side comparison of classical vs. quantum-enhanced results

A novel contribution to cybersecurity research in Pakistan · Achievable within a 4-month timeline · Open-source tools

TOOLS & TECHNOLOGIES

Technical Stack

All tools are free, open-source, and well-documented

Core Language

- Python 3.x
- Jupyter Notebook
- VS Code IDE

ML Libraries

- scikit-learn
- TensorFlow / Keras
- NumPy / Pandas

Quantum (QPSO)

- DEAP library
- pyswarm
- Custom QPSO impl.

Visualization

- Matplotlib
- Seaborn
- Plotly

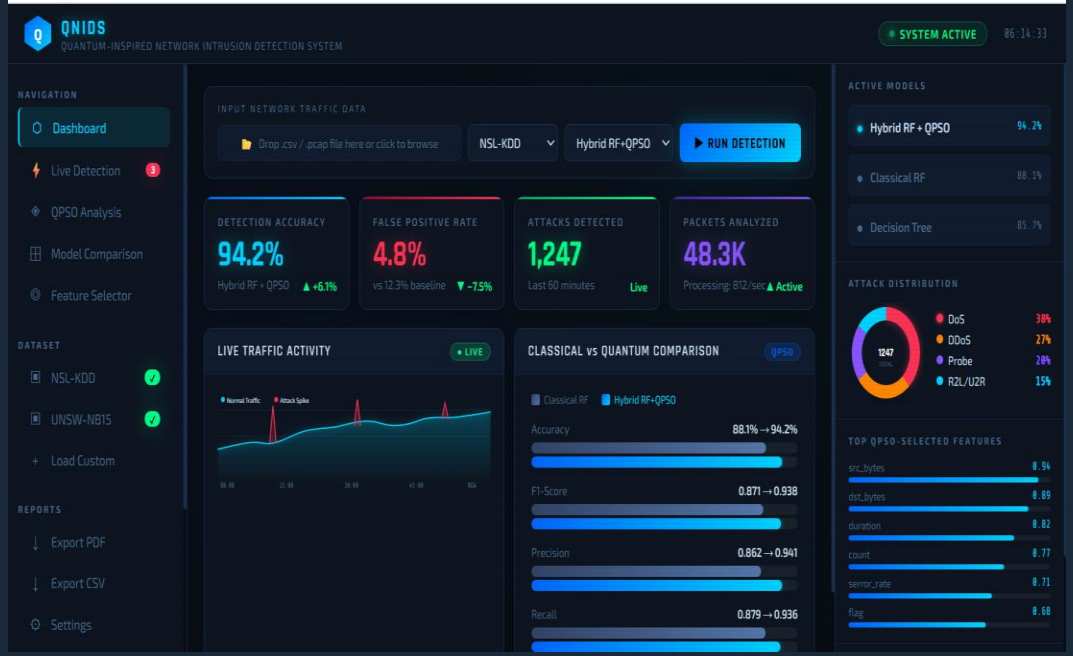
GUI Framework

- Tkinter / PyQt5
- Dark dashboard
- Live chart embed

Datasets

- NSL-KDD (UNB.ca)
- UNSW-NB15
- Publicly available

Expected Interface



Supervised by: Prof. Dr Syed Ali Abbas | BSCS 2022–2026